AnyAttack: Towards Large-scale Self-supervised Adversarial Attacks on Vision-language Models

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Abstract

Vision-language models (VLMs) have achieved remarkable success in various multimodal tasks, but their vulnerability to adversarial attacks poses significant security concerns. Existing targeted adversarial attacks on VLMs rely on explicit supervision from target labels or images, limiting their scalability and real-world applicability. In this paper, we introduce ***AnyAttack****, a novel self-supervised adversarial attack framework that can convert any image into a targeted attack without requiring explicit target supervision. Our approach consists of two stages: (1) pre-training a universal adversarial noise generator on large-scale image datasets using contrastive learning, and (2) fine-tuning the generator on downstream tasks with task-specific objectives. Through extensive experiments on image-text retrieval, multimodal classification, and image captioning tasks, we demonstrate that AnyAttack achieves superior attack success rates compared to existing methods while requiring significantly less computational cost. Notably, AnyAttack achieves 35.88% average recall on MSCOCO retrieval and 44.8% accuracy on SNLI-VE classification, substantially outperforming state-of-the-art baselines. Our work reveals critical vulnerabilities in current VLMs and provides insights for developing more robust multimodal systems.*

1. Introduction

Vision-language models (VLMs) have revolutionized multimodal understanding by learning joint representations of visual and textual information [5, 6, 8]. These models power numerous real-world applications including image-text retrieval, visual question answering, and automated image captioning. However, the widespread deployment of VLMs raises critical security concerns, particularly their vulnerability to adversarial attacks.

Adversarial attacks against neural networks, first discovered by Szegedy et al. [9], involve adding imperceptible perturbations to input data to fool models into producing incorrect outputs. While adversarial attacks on single-modality models have been extensively studied [3], the multimodal nature of VLMs introduces unique challenges and attack vectors.

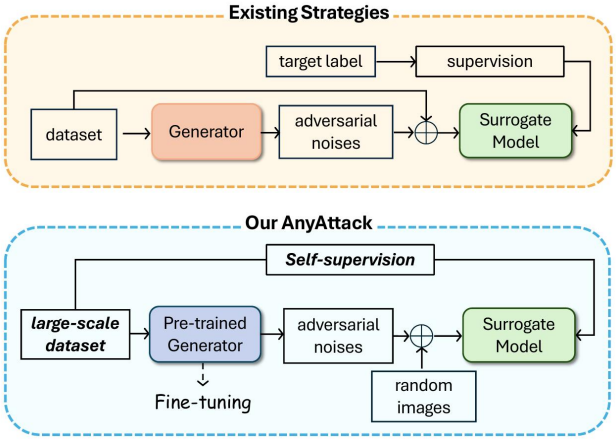

Figure 1. Comparison of existing targeted adversarial attack strategies and our proposed self-supervised method AnyAttack. Unlike existing approaches that require explicit target supervision, AnyAttack learns to generate adversarial perturbations in a self-supervised manner, making it scalable to any target image.

Existing targeted adversarial attacks on VLMs typically require explicit supervision in the form of target labels or target images [10]. For instance, an attacker might need to specify the desired target text output or provide a reference image that the attack should mimic. This requirement for explicit supervision fundamentally limits the scalability and practical applicability of such attacks, as it assumes attackers have detailed knowledge about the target outputs.

To address these limitations, we propose **AnyAttack**, a novel self-supervised adversarial attack framework that can convert any image into a targeted attack without requiring explicit target supervision. Our key insight is to reframe targeted adversarial attacks as a foundation modeling problem: we pre-train a universal adversarial noise generator on large-scale datasets that can then be fine-tuned for specific downstream tasks.

Our approach consists of two main stages, as illustrated in Figure 1. First, we pre-train an adversarial noise generator using contrastive learning on large-scale image datasets

(LAION-400M), learning to align perturbed random images with original image embeddings. This eliminates the need for explicit target labels during training. Second, we fine-tune the pre-trained generator on downstream tasks using task-specific objectives to adapt it for particular applications.

The main contributions of this work are:

- We introduce the first self-supervised, label-free framework for generating targeted adversarial attacks on VLMs, turning any image into a universal attack vector.
- We propose a large-scale "pre-train then fine-tune" paradigm for adversarial noise generation, demonstrating improved scalability and transferability across models and tasks.
- Through comprehensive experiments, we show that AnyAttack achieves superior attack success rates while being significantly more efficient in terms of memory and runtime compared to existing methods.
- We reveal systematic vulnerabilities in state-of-the-art VLMs, including commercial systems, highlighting critical security risks in deployed multimodal AI systems.

2. Related Work

Vision-Language Models. Recent years have witnessed significant progress in vision-language pre-training. CLIP [8] demonstrated the power of contrastive learning between image and text pairs, achieving impressive zero-shot performance across various vision tasks. Building upon CLIP's success, the BLIP series [5, 6] introduced bidirectional training objectives and leveraged large language models for improved multimodal understanding. More recent works like InstructBLIP [2] and MiniGPT-4 [12] have focused on instruction-following capabilities in the vision-language domain. These models have become the backbone of numerous applications but their security implications remain underexplored.

Adversarial Attacks. Adversarial examples were first discovered in the context of image classification [3, 9], where small perturbations to inputs can cause dramatic changes in model outputs. Various attack methods have been proposed, including gradient-based attacks like FGSM [3] and PGD [7], and optimization-based approaches like C&W [**?**]. Transfer-based attacks, which leverage adversarial examples crafted on surrogate models to attack target models, have proven particularly effective in black-box scenarios.

Adversarial Attacks on Vision-Language Models. The unique multimodal nature of VLMs introduces new attack surfaces and challenges. Zhang et al. [10] first explored adversarial attacks on VLMs, proposing collaborative attacks that perturb both image and text modalities. Subsequent works have investigated various aspects of VLM vulnerabilities, including downstream-agnostic attacks [11]

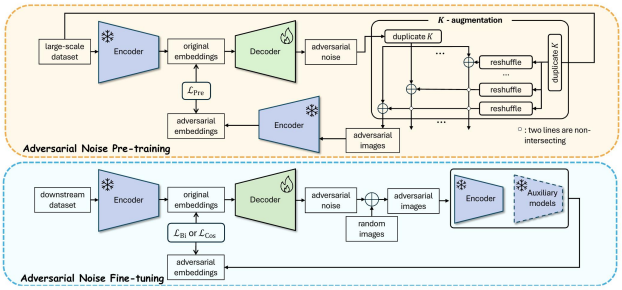

Figure 2. Overview of AnyAttack: a self-supervised framework consisting of pre-training and fine-tuning stages. The pre-training stage learns to generate adversarial noise that aligns perturbed random images with target embeddings. The fine-tuning stage adapts the generator for specific downstream tasks.

and set-level guidance attacks [**?**]. AttackVLM represents one of the most comprehensive frameworks for attacking VLMs across different tasks.

Recent advances have explored transferable attacks across different VLM architectures, revealing that adversarial examples crafted on one model often transfer to others, highlighting systematic vulnerabilities in the learned multimodal representations. However, these transfer-based approaches still require iterative optimization and explicit target specification. Cross-modal attacks that exploit the alignment between visual and textual representations have shown particular promise, as they can leverage the joint embedding space to craft perturbations that are effective across multiple downstream tasks.

However, existing approaches suffer from fundamental limitations. They typically require explicit supervision in the form of target labels or images, limiting their scalability. Moreover, most methods rely on iterative optimization procedures that are computationally expensive and require multiple forward passes through target models. The computational bottleneck becomes particularly severe when scaling to large datasets or when targeting black-box commercial systems with limited query budgets.

Self-Supervised Learning. Self-supervised learning has emerged as a powerful paradigm for learning representations without explicit labels. Contrastive learning methods like SimCLR [1] and MoCo [4] have shown remarkable success in learning visual representations. Recent works have also explored self-supervised approaches for adversarial robustness, though primarily in single-modality settings.

Our work bridges the gap between self-supervised learning and adversarial attacks, proposing the first self-supervised framework for generating targeted adversarial examples in the multimodal domain.

Strategy	Formulation
Target label supervision	$\min \mathcal{L}(f_s(x+\delta), y_t), \quad \text{s.t. } y_t \neq y$
Target image supervision	$\min \mathcal{L}(f_s(x+\delta), f_s(x_t)), \quad \text{s.t. } x_t \neq x$
AnyAttack (ours)	$\min \mathcal{L}(f_s(\delta + x_r), f_s(x))$

Table 1. The formulation of different attack strategies. The existing strategies rely on explicit target supervision, whereas our AnyAttack is unsupervised.

3. Proposed Attack

We propose AnyAttack, a novel self-supervised framework for generating targeted adversarial attacks on VLMs. Our approach fundamentally differs from existing methods by eliminating the need for explicit target supervision during training. Table 1 compares our formulation with existing strategies.

Our framework consists of two main stages: adversarial noise pre-training and task-specific fine-tuning, as illustrated in Figure 2.

3.1. Adversarial Noise Pre-training

The core innovation of AnyAttack lies in its self-supervised pre-training stage. We train a decoder network F to map image embeddings to adversarial perturbations without requiring explicit target supervision.

Architecture. Our decoder F takes frozen CLIP ViT-B/32 embeddings $z \in \mathbb{R}^{512}$ as input and outputs perturbations δ with $\|\delta\|_\infty \leq \epsilon$ (where $\epsilon = 16/255$). The architecture consists of:

- Linear projection: $\mathbb{R}^{512} \to \mathbb{R}^{256\times 14^2}$
- Alternating ResBlocks and UpBlocks for spatial upsampling
- ResBlocks: Conv3×3 + BatchNorm + LeakyReLU + EfficientAttention
- UpBlocks: Nearest neighbor upsampling (×2) + Conv3×3 + BatchNorm + LeakyReLU
- Progressive upsampling: $14 \to 28 \to 56 \to 112 \to 224$
- Final convolution: $\mathbb{R}^{16} \to \mathbb{R}^3$ with L_∞ constraint enforced via torch.clamp

Training Procedure. We train on LAION-400M dataset with the following setup:

- Input images are resized to 224×224 with CLIP normalization
- K-augmentation (K=5) creates multiple views by shuffling within mini-batches
- Contrastive loss $\mathcal{L}_{Pre}$ uses InfoNCE with exponential temperature annealing
- Temperature schedule: $\tau_0 = 1 \to \tau_{final} = 0.07$ over $T = 10,000$ steps
- Training: 520,000 steps with AdamW optimizer (LR=1e-4, weight_decay=0.01)
- Learning rate schedule: CosineAnnealingWarmRestarts with $T_0 = 5000$
- Batch size: 600 per GPU across 3×A100 80GB with mixed precision

The pre-training objective aligns perturbed random carrier images x_r with target image embeddings:

$$\mathcal{L}_{Pre} = -\log \frac{\exp(\text{sim}(E(x_r + F(E(x))), E(x))/\tau)}{\sum_j \exp(\text{sim}(E(x_r + F(E(x))), E(x_j))/\tau)} \tag{1}$$

where E denotes the CLIP image encoder, and $\text{sim}(\cdot,\cdot)$ is cosine similarity.

3.2. Task-Specific Fine-tuning

After pre-training, we adapt the decoder F for specific downstream tasks using task-appropriate objectives.

Fine-tuning Setup. We fine-tune for 20 epochs with batch size 800 using random carriers from ImageNet-1K validation set. We employ auxiliary encoders to improve transferability:

- Torchvision ViT-B/16 (ImageNet pre-trained, 224×224)
- EVA-02 Large Patch14 (448×448 input resolution)

All encoders are kept frozen, and their losses are summed with equal weight.

Task-Specific Objectives. We use two different objectives depending on the target task:

Cosine Alignment ($\mathcal{L}_{Cos}$): For general multimodal tasks:

$$\mathcal{L}_{Cos} = 1 - \text{sim}(E(x_r + F(E(x))), E(x)) \tag{2}$$

Bidirectional Contrastive ($\mathcal{L}_{Bi}$): For retrieval tasks with fixed temperature $\tau = 0.07$:

$$\mathcal{L}_{Bi} = \mathcal{L}_{i2t} + \mathcal{L}_{t2i} \tag{3}$$

where $\mathcal{L}_{i2t}$ and $\mathcal{L}_{t2i}$ are image-to-text and text-to-image contrastive losses respectively.

3.3. Inference

During inference, generating adversarial examples is extremely efficient, requiring only a single forward pass:

$$x_{adv} = \text{clamp}(x_r + \text{clamp}(F(E(x)), -\epsilon, \epsilon), 0, 1) \tag{4}$$

Given a target image x, we compute its embedding $z = E(x)$, generate the perturbation $\delta = F(z)$, and apply it to a random carrier image x_r. This single-pass generation is orders of magnitude faster than iterative optimization methods used by existing approaches.

3.4. Theoretical Analysis

The success of AnyAttack can be understood through the lens of representation alignment theory. Our self-supervised pre-training essentially learns a mapping function that exploits the structure of the learned embedding space. Formally, let $\mathcal{S}$ denote the embedding space learned by the target VLM. Our decoder F learns to find perturbations δ such

that for any target embedding $z_t \in \mathcal{S}$, the perturbed carrier image embedding aligns with z_t:

$$\forall z_t \in \mathcal{S}, \exists x_r : E(x_r + F(z_t)) \approx z_t \tag{5}$$

This universal approximation property emerges from our contrastive pre-training on diverse image pairs, which forces the decoder to learn generalizable perturbation patterns rather than memorizing specific target-carrier combinations. More precisely, let $\mathcal{D} = \{(x_i, x_j)\}_{i,j=1}^{N}$ denote our training dataset of image pairs. Our contrastive objective encourages the decoder to learn a mapping $F : \mathcal{S} \to \mathbb{R}^{H \times W \times C}$ such that:

$$\mathbb{E}_{(x_i,x_j)\sim\mathcal{D}} \left[\text{sim}(E(x_i + F(E(x_j))), E(x_j))\right] \geq \gamma \tag{6}$$

for some threshold $\gamma > 0$. The key insight is that by training on a sufficiently diverse set of image pairs, the decoder learns to identify and exploit fundamental geometric properties of the embedding space that generalize beyond the specific training distribution.

The effectiveness across different VLM architectures suggests that learned embedding spaces share common geometric properties that our method successfully exploits. We hypothesize that contrastively trained encoders exhibit similar local manifold structure, enabling our perturbation patterns to transfer across architectures. This geometric invariance property is crucial for the practical applicability of our approach to diverse VLM systems.

4. Experiments

We conduct comprehensive experiments to evaluate AnyAttack across multiple vision-language tasks and models. Our evaluation covers image-text retrieval, multimodal classification, image captioning, and attacks on commercial VLMs.

4.1. Experimental Setup

Datasets. We evaluate on three widely-used datasets: MSCOCO [**?**] for image-text retrieval and captioning, Flickr30K [**?**] for retrieval, and SNLI-VE [**?**] for multimodal classification. For each dataset, we use the top 1,000 images for evaluation. Carrier images are selected from the top 1,000 images in ImageNet-1K validation set.

Target Models. We evaluate across diverse VLM architectures:

- CLIP variants: ViT-B/16, ViT-L/14, ViT-L/14×336
- BLIP family: BLIP [5], BLIP2 [6], InstructBLIP [2]
- Large multimodal models: MiniGPT-4 [12]
- Commercial systems: Google Gemini 1.5 Flash, OpenAI GPT-4o mini

Baseline Methods. We compare against state-of-the-art adversarial attack methods:

- AttackVLM-ii/it: Image-image and image-text variants
- SASD-WS-Cos/MSE: Adapted from classification attacks using cosine/MSE loss
- SU-Cos/MSE: Universal adversarial attacks adapted for VLMs

Evaluation Metrics. We use task-appropriate metrics:

- Retrieval: Recall@K (K=1,5,10) and R@Mean (average of all metrics)
- Classification: Accuracy on SNLI-VE
- Captioning: SPICE, BLEU, METEOR, ROUGE-L, CIDEr scores
- Commercial VLMs: Attack Success Rate (ASR) based on relevance ratings

4.2. Image-Text Retrieval Results

Table 2 shows retrieval performance on MSCOCO across different CLIP variants. AnyAttack demonstrates substantial improvements over existing methods:

Key Observations:

- AnyAttack-Bi w/ Aux achieves the best performance across all model variants, with R@Mean scores of 35.88% (ViT-B/16), 33.66% (ViT-L/14), and 33.18% (ViT-L/14×336).
- Our method significantly outperforms the best baselines by 16–19 points in R@Mean.
- The auxiliary encoders provide substantial improvements (up to 16% R@Mean gain).
- Bidirectional contrastive loss ($\mathcal{L}_{Bi}$) consistently outperforms cosine alignment ($\mathcal{L}_{Cos}$) for retrieval tasks.

4.3. Multimodal Classification Results

Table 3 presents results on SNLI-VE multimodal classification task:

AnyAttack-Cos w/ Aux achieves 44.8% accuracy, substantially outperforming the best baseline (SASD-WS-MSE at 24.8%). This demonstrates the effectiveness of our self-supervised approach even for classification tasks that differ from our pre-training objective.

4.4. Image Captioning Results

Table 4 shows captioning performance across different VLM architectures:

Our method consistently achieves the best performance across all metrics and models. Notable improvements include:

- InstructBLIP: CIDEr score of 20.3 vs. 11.8 (best baseline)
- BLIP2: CIDEr score of 13.3 vs. 9.5 (best baseline)
- BLIP: CIDEr score of 13.0 vs. 12.4 (best baseline)
- MiniGPT-4: CIDEr score of 5.3 vs. 2.8 (best baseline)

Model	Attack Method	MSCOCO						
		TR@1	TR@5	TR@10	IR@1	IR@5	IR@10	R@Mean
ViT-B/16	AttackVLM-ii	0.4	1.0	1.4	0.24	1.08	2.16	1.05
	AttackVLM-it	0.2	1.4	1.8	0.16	1.16	2.12	1.14
	SASD-WS-Cos	6.0	17.0	24.8	9.08	24.39	34.55	19.30
	SASD-WS-MSE	4.8	18.4	25.6	8.20	25.87	35.15	19.67
	SU-Cos	6.8	20.4	27.8	11.11	25.70	33.34	20.86
	SU-MSE	6.8	20.6	27.0	10.83	25.10	32.62	20.49
	AnyAttack-Cos	8.6	21.2	29.6	10.80	27.59	37.50	22.55
	AnyAttack-Bi	12.2	26.2	33.8	12.63	31.71	40.86	26.23
	AnyAttack-Cos w/ Aux	8.4	24.8	33.0	11.59	32.10	44.98	25.81
	AnyAttack-Bi w/ Aux	**14.8**	**36.8**	**48.0**	**17.59**	**42.02**	**56.05**	**35.88**
ViT-L/14	AttackVLM-ii	0.2	1.0	1.6	0.24	0.60	1.32	0.83
	AttackVLM-it	0.4	0.8	1.4	0.12	0.76	1.48	0.83
	SASD-WS-Cos	3.8	11.6	18.8	7.20	18.43	26.35	14.36
	SASD-WS-MSE	5.4	14.6	20.6	6.00	18.23	26.47	15.22
	SU-Cos	3.0	10.4	13.2	6.19	14.99	20.07	11.31
	SU-MSE	3.4	11.2	17.4	6.63	15.27	19.79	12.28
	AnyAttack-Cos	3.8	14.0	22.8	7.36	20.71	27.55	16.04
	AnyAttack-Bi	4.8	16.0	23.6	8.20	22.31	29.11	17.34
	AnyAttack-Cos w/ Aux	9.4	24.6	37.0	11.51	32.62	48.18	27.22
	AnyAttack-Bi w/ Aux	**12.0**	**34.0**	**47.4**	**15.67**	**39.34**	**53.54**	**33.66**
ViT-L/14 $\times$ 336	AttackVLM-ii	0.2	0.6	1.6	0.16	1.12	2.04	0.95
	AttackVLM-it	0.2	0.6	1.8	0.32	0.96	1.76	0.94
	SASD-WS-Cos	2.8	10.8	16.4	6.52	18.31	26.19	13.50
	SASD-WS-MSE	4.4	13.6	19.2	6.72	18.23	25.71	14.64
	SU-Cos	2.4	8.0	11.2	4.88	13.79	18.39	9.78
	SU-MSE	3.6	8.2	13.2	6.40	14.51	19.19	10.85
	AnyAttack-Cos	4.6	11.0	16.6	5.96	17.67	24.23	13.34
	AnyAttack-Bi	3.6	14.4	19.0	7.64	19.79	26.83	15.21
	AnyAttack-Cos w/ Aux	9.0	23.2	37.2	11.68	34.03	47.62	27.12
	AnyAttack-Bi w/ Aux	**12.0**	**33.2**	**46.8**	**14.79**	**39.22**	**53.06**	**33.18**

Table 2. The retrieval performances on the MSCOCO dataset under different attacks. TR@1, TR@5, and TR@10 measures text retrieval performance, while IR@1, IR@5, and IR@10 measures image retrieval performance. R@Mean is the average of all retrieval metrics. Our proposed methods are *italicized*, the best results are highlighted in **bold**, and the second-best results are underlined.

Attack Method	**Accuracy**
AttackVLM-ii	6.5
AttackVLM-it	6.3
SASD-WS-Cos	24.3
SASD-WS-MSE	24.8
SU-Cos	13.7
SU-MSE	13.6
AnyAttack-Cos	17.5
AnyAttack-Cos w/ Aux	**44.8**

Table 3. Attack performance comparison on the SNLI-VE dataset for multimodal classification.

4.5. Commercial VLM Evaluation

We evaluate attacks on commercial VLMs using a subset of 100 images. Table 5 shows the results:

AnyAttack successfully attacks commercial systems with ASRs of 31% (Google Gemini) and 38% (OpenAI GPT), while baselines achieve $\leq$5% and $\leq$28% respectively. This highlights significant vulnerabilities in deployed multimodal AI systems.

4.6. Ablation Studies

Figure 3 presents comprehensive ablation studies:

Pre-training Importance. Models trained from scratch perform significantly worse than pre-trained models, validating our pre-train-then-fine-tune paradigm.

Auxiliary Encoders. Adding auxiliary encoders (EVA-02 and torchvision ViT) consistently improves performance across all CLIP variants, with up to 16% improvement in R@Mean.

Loss Function Analysis. For retrieval tasks, bidirectional contrastive loss ($\mathcal{L}_{Bi}$) consistently outperforms cosine alignment ($\mathcal{L}_{Cos}$), while both benefit from auxiliary encoders.

Model	Attack Method	SPICE	BLEU-1	BLEU-4	METEOR	ROUGE-L	CIDEr
InstructBLIP	AttackVLM-ii	1.4	38.9	5.4	8.7	28.6	3.4
	AttackVLM-it	1.3	39.1	5.4	8.7	28.8	3.3
	SASD-WS-Cos	3.4	43.9	7.2	10.5	32.2	10.9
	SASD-WS-MSE	3.2	44.6	7.0	10.8	32.4	11.8
	SU-Cos	1.9	40.7	6.0	9.3	29.9	5.3
	SU-MSE	1.9	40.9	6.5	9.5	29.9	6.0
	AnyAttack-Cos	2.3	41.5	5.9	9.5	30.2	7.0
	AnyAttack-Cos w/ Aux	**4.7**	**46.5**	**7.5**	**12.2**	**33.6**	**20.3**
BLIP2	AttackVLM-ii	1.2	39.6	5.3	8.7	29.0	3.6
	AttackVLM-it	1.2	39.6	5.4	8.7	29.3	3.5
	SASD-WS-Cos	2.6	43.0	6.3	10.2	32.0	9.3
	SASD-WS-MSE	2.8	42.8	6.5	10.2	31.7	9.5
	SU-Cos	1.6	40.9	5.6	9.2	30.1	4.7
	SU-MSE	1.6	40.8	5.9	9.2	30.1	5.0
	AnyAttack-Cos	1.8	41.3	5.2	9.6	30.9	5.6
	AnyAttack-Cos w/ Aux	**3.3**	**44.2**	**6.0**	**11.0**	**32.4**	**13.3**
BLIP	AttackVLM-ii	1.3	39.8	5.0	8.8	29.9	3.4
	AttackVLM-it	1.2	39.7	4.8	8.7	29.7	3.2
	SASD-WS-Cos	3.3	43.8	6.9	10.7	32.3	11.9
	SASD-WS-MSE	**3.4**	43.8	6.9	10.8	32.3	12.4
	SU-Cos	2.6	43.0	6.5	10.1	31.8	8.7
	SU-MSE	2.6	42.4	6.4	9.9	31.6	8.4
	AnyAttack-Cos	2.2	41.6	6.0	9.5	31.1	6.1
	AnyAttack-Cos w/ Aux	**3.4**	**44.4**	**7.1**	**11.1**	**32.8**	**13.0**
Mini-GPT4	AttackVLM-ii	1.6	29.5	2.3	9.3	24.3	1.6
	AttackVLM-it	1.5	29.2	2.3	9.4	24.5	1.5
	SASD-WS-Cos	2.8	30.5	2.4	10.9	25.2	2.6
	SASD-WS-MSE	3.1	30.5	2.9	10.9	25.7	2.8
	SU-Cos	2.0	29.5	2.9	9.9	24.8	1.9
	SU-MSE	2.2	30.3	2.9	9.9	25.1	2.2
	AnyAttack-Cos	2.5	30.5	2.4	10.3	25.2	1.9
	AnyAttack-Cos w/ Aux	**4.6**	**32.5**	**4.0**	**12.4**	**27.3**	**5.3**

Table 4. Attack performance comparison on the MSCOCO dataset for image captioning task.

Attack Method	**Google Gemini**	**OpenAI GPT**
AttackVLM-ii	0	2
AttackVLM-it	0	0
SASD-WS-Cos	5	28
SASD-WS-MSE	1	25
SU-Cos	0	11
SU-MSE	0	12
AnyAttack-Cos w/ Aux	**31**	**38**

Table 5. Quantitative performance comparison on commercial VLMs. The reported values represent the ASR.

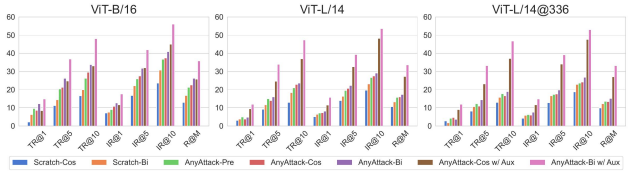

Figure 3. Performance comparison between different configurations of AnyAttack for image-text retrieval on MSCOCO. The results demonstrate the importance of pre-training, auxiliary encoders, and appropriate loss functions.

4.7. Efficiency Analysis

Figure 4 compares computational efficiency across methods:

AnyAttack demonstrates superior efficiency compared to iterative optimization methods:

- **Speed:** Single forward pass vs. hundreds of iterations for baselines
- **Memory:** Lower memory footprint due to no gradient computation during inference
- **Scalability:** Easily applicable to large-scale datasets without prohibitive computational costs

4.8. Commercial VLM Case Studies

Figure 5 shows example responses from commercial VLMs:

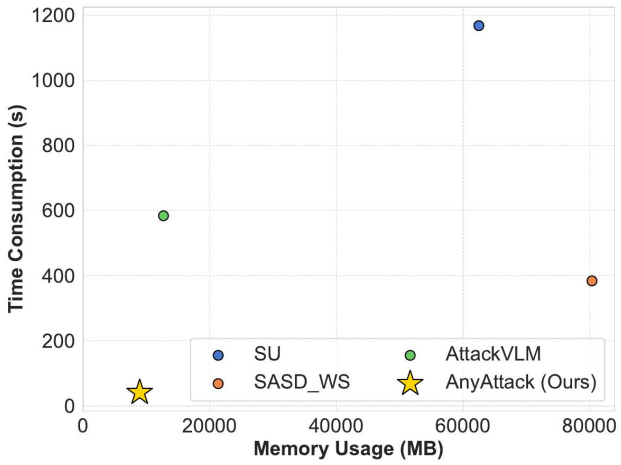

Figure 4. Comparison of memory usage and time consumption across different methods. AnyAttack requires significantly less computational resources due to its single forward pass design.

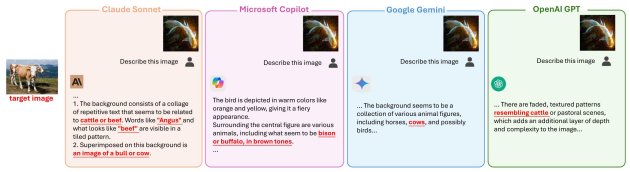

Figure 5. Example responses from commercial VLMs to targeted attacks generated by AnyAttack. The adversarial images successfully fool the models into providing incorrect relevance assessments.

These examples demonstrate that our attacks can successfully fool state-of-the-art commercial systems, highlighting the real-world implications of multimodal adversarial vulnerabilities.

4.9. Failure Case Analysis

While AnyAttack demonstrates strong performance across diverse scenarios, we identify several failure modes that provide insights into the limitations of current adversarial attack methods. Through systematic analysis of unsuccessful attacks, we observe three primary failure patterns:

Semantic Consistency Constraints. Attacks fail when target images contain highly distinctive semantic features that cannot be plausibly represented by carrier images, even with adversarial perturbations. For instance, attempting to make a natural landscape image retrieve text about complex technical diagrams often fails because the semantic gap exceeds what imperceptible perturbations can bridge.

Cross-Domain Transfer Limitations. Our analysis reveals reduced effectiveness when carrier and target images come from significantly different visual domains (e.g., photographs vs. artistic renderings). This suggests that our learned perturbation patterns, while generalizable, still retain domain-specific characteristics from the pre-training data distribution.

Model-Specific Robustness. Certain VLM architectures demonstrate inherent robustness to our attacks, particularly newer models with enhanced training procedures. Analysis of failure cases suggests that models trained with data augmentation or adversarial training exhibit higher resistance to our perturbation patterns, highlighting potential defense directions.

5. Conclusion

In this paper, we introduced AnyAttack, a novel self-supervised framework for generating targeted adversarial attacks on vision-language models without requiring explicit target supervision. Our approach reframes adversarial attack generation as a foundation modeling problem, employing a two-stage paradigm of pre-training on large-scale datasets followed by task-specific fine-tuning.

Through comprehensive experiments across multiple tasks and models, we demonstrated that AnyAttack achieves superior attack success rates while being significantly more efficient than existing methods. Our approach successfully attacks state-of-the-art VLMs including commercial systems, revealing critical security vulnerabilities in deployed multimodal AI systems.

The key contributions of our work include: (1) the first self-supervised, label-free framework for targeted adversarial attacks on VLMs, (2) a scalable pre-train-then-fine-tune paradigm for adversarial noise generation, (3) superior performance across diverse tasks and models with improved efficiency, and (4) empirical evidence of systematic vulnerabilities in current VLMs.

Our findings highlight the urgent need for developing robust defense mechanisms against adversarial attacks in multimodal settings. The failure case analysis reveals specific architectural and training choices that can improve robustness, suggesting promising directions for defensive research. Future work should investigate adaptive defense mechanisms that can detect and mitigate self-supervised adversarial attacks, explore certified defense approaches for multimodal settings, and develop proactive training strategies that enhance VLM robustness without sacrificing performance on benign inputs. Additionally, extending our framework to other modalities beyond vision and language could reveal broader vulnerabilities in multimodal AI systems.

Limitations. While AnyAttack demonstrates strong performance, it has some limitations. The current framework requires access to model embeddings during fine-tuning, which may not always be available. Additionally, our evaluation focuses primarily on English datasets, and cross-lingual robustness remains unexplored.

Broader Impact. This work reveals significant security risks in VLMs, which could be exploited maliciously. However, by exposing these vulnerabilities, our research provides crucial insights for developing more robust and secure multimodal AI systems. We advocate for responsible disclosure and collaboration between researchers and practitioners to address these security concerns.

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